LocalRAG Architecture and Performance Guide

Overview:

LocalRAG is a microservices-based Retrieval-Augmented Generation application designed to run entire
It uses Ollama to host both the generation language model (Qwen 2.5 0.5B) and embedding model (No
The vector store is ChromaDB backed by SQLite for metadata tracking.

Telemetry and Observability:

All operations in LocalRAG emit metrics to Prometheus Pushgateway on port 9091.
Prometheus Server scrapes Pushgateway on port 9091 every 5 seconds.
Grafana displays real-time telemetry dashboards on port 3001 with Prometheus as its datasource.

Host CLI:

A rich command line interface built with Typer and Rich allows administrators to ingest documents, run queries, launch interactive chat, and check system health directly from the host terminal.